

Perturbation Catalogue

An Integrated Resource for Exploring the Impact of Genetic Perturbations

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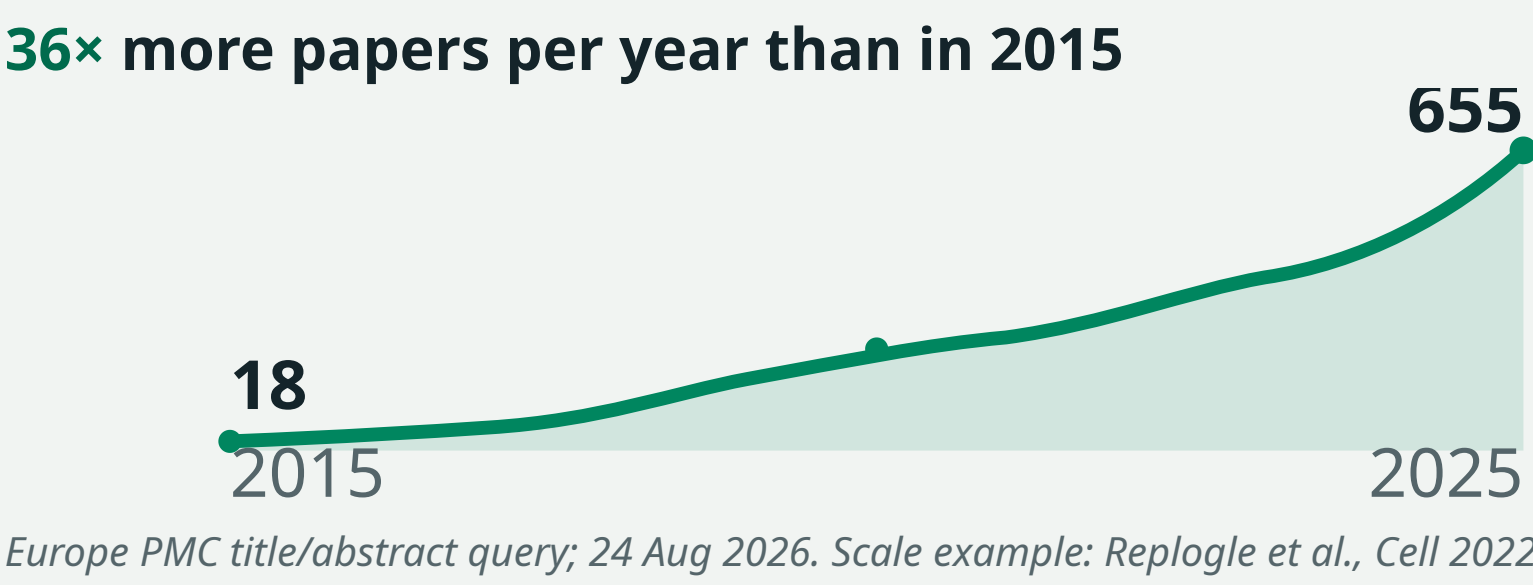
Wellcome Genome

Campus

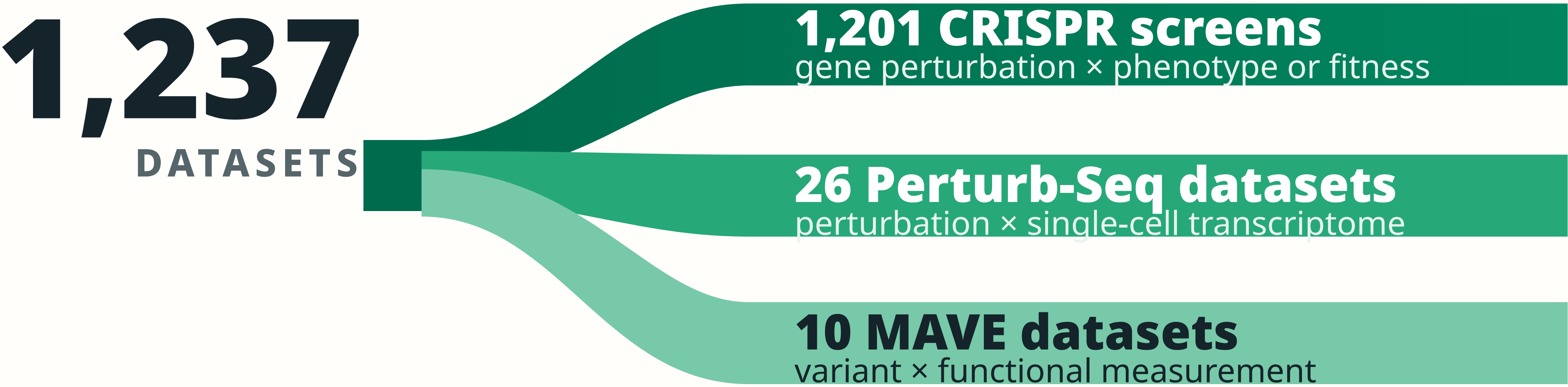
Hinxton, United Kingdom

Perturbation evidence is multiplying faster than it can be organised

A genome-scale Perturb-Seq study can profile **more than 2.5 million cells**. But studies arrive with different identifiers, metadata and processing choices, so someone still has to reconstruct the experimental context before results can be compared.



A single catalogue turns scattered evidence into a searchable resource



19,235

human targets

36

tissues

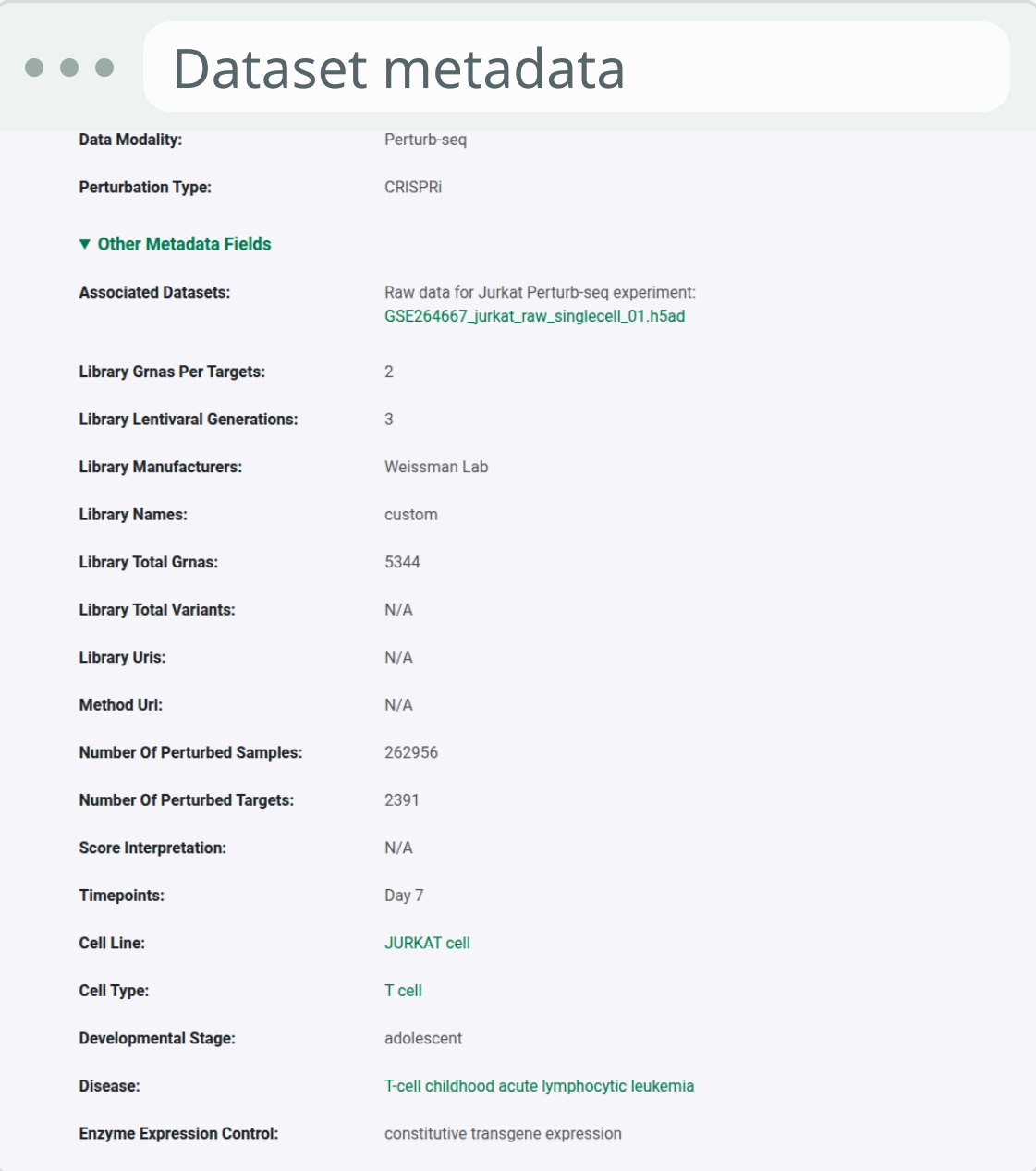
34

cell types

1,200

cell lines

Metadata and reanalysis make results comparable



One metadata contract

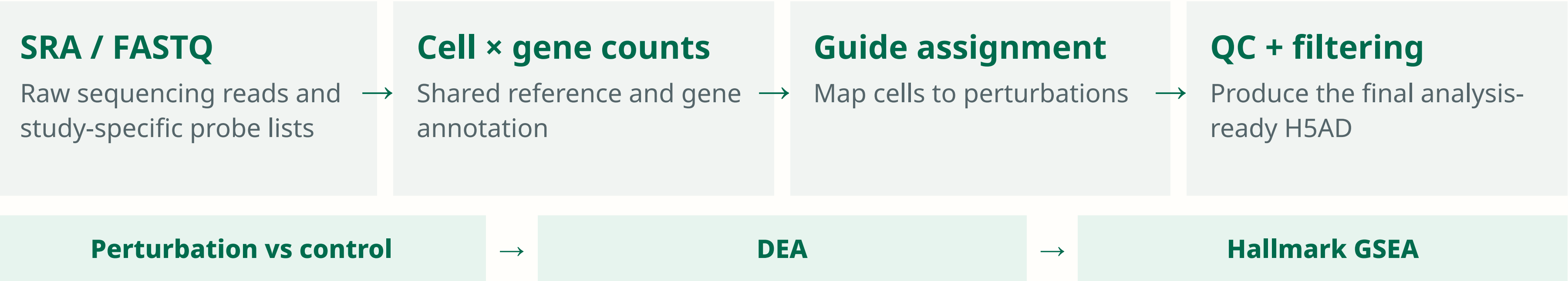
A validated, extensible schema records the experiment needed to interpret a score—not merely its accession.

Study · perturbation · assay · model system · tissue · cell type or line · disease · timepoint · licence · provenance

Ensembl gene IDs provide stable target identity; biological entities link to established ontologies.

Perturb-Seq reprocessing starts again from raw reads

Author matrices differ in alignment, annotation, filtering and guide assignment. Selected studies follow one shared computational path.



Five datasets are live. Broad agreement with author-generated matrices is checked where available; independent quality metrics and wider kit support are next.

Every result is programmatically accessible

Search, metadata, modality results and complete downloads use the same production backend as the portal.

```
GET /search?query=BRCA2&search_mode=targets
GET /dataset/nadig_2025_jurkat
GET /v1/perturb-seq/nadig_2025_jurkat/search?limit=1
GET /v1/perturb-seq/nadig_2025_jurkat/download?format=csv.gz
```

```
{
  "total_rows_count": 30078802,
  "results": [{
    "perturbation": {
      "gene_symbol": "TFAM",
      "perturbed_target_ensg": "ENSG00000108064"
    },
    "effect": {
      "gene_symbol": "MT-ND5",
      "log2fc": -1.926,
      "padj": 0.0,
      "direction": "decreased"
    }
  }]
}
```

Next, integration becomes continuous and the analytical foundation becomes stronger

Extract structured metadata faster while retaining curator review

AI-assisted drafting, ambiguity resolution and validated ontology mapping before ingestion.

Reanalyse more designs and measure quality independently

More kits and experimental structures, author-independent QC and modality-appropriate methods.

Publish periodic, versioned releases with explicit provenance

Stable snapshots and transparent analytical changes for reproducible downstream use.

Bring us the next hard perturbation-data problem

Suggest datasets, challenge the schema or develop the resource with us. We welcome scientific partnerships and funded multi-year collaborations around your programme's modalities, analyses and functionality.

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Explore the Catalogue

EMBL-EBI

Open Targets